



Lecture 8 – Introduction to Machine Learning

30 Exam-Style MCQs



Part 1: Fundamentals of Machine Learning

1.

Machine Learning is best described as a field that:

- A. Requires explicit programming of all rules
- B. Allows computers to learn patterns from data without being explicitly programmed
- C. Focuses only on hardware acceleration
- D. Eliminates the need for data

Answer: B

2.

According to Arthur Samuel (1959), machine learning gives computers the ability to:

- A. Execute faster programs
- B. Learn without being explicitly programmed
- C. Replace software engineers
- D. Optimize memory usage

Answer: B

3.

In traditional programming, the programmer provides:

- A. Data and output only
- B. Data and rules (program)

- C. Output and learned model
- D. Training labels only

Answer: B

4.

In machine learning, the system is trained using:

- A. Rules and expected execution paths
- B. Only unlabeled data
- C. Data and desired outputs
- D. Compiler optimizations

Answer: C

5.

The “lazy way” to solve character recognition refers to:

- A. Writing heuristic rules
- B. Using hard-coded templates
- C. Training a model using large amounts of data
- D. Ignoring accuracy

Answer: C



Part 2: AI vs ML vs Deep Learning

6.

Machine Learning is best described as a subset of:

- A. Software Engineering
- B. Databases
- C. Artificial Intelligence
- D. Computer Networks

Answer: C

7.

Deep Learning is characterized mainly by:

- A. Rule-based reasoning
- B. Shallow models only
- C. Neural networks with multiple layers
- D. Manual feature extraction

Answer: C



Part 3: Learning Types

8.

Supervised learning requires training data that:

- A. Contains no output labels
- B. Contains only inputs
- C. Contains input–output pairs
- D. Contains rewards only

Answer: C

9.

Which of the following is a supervised learning task?

- A. Clustering
- B. Dimensionality reduction
- C. Classification
- D. Feature extraction

Answer: C

10.

Regression differs from classification because its output is:

- A. Discrete
- B. Binary

- C. Continuous
- D. Unlabeled

Answer: C

11.

Unsupervised learning aims primarily to:

- A. Predict known outputs
- B. Learn structure from unlabeled data
- C. Maximize reward
- D. Minimize runtime

Answer: B

12.

Which task belongs to unsupervised learning?

- A. Spam detection
- B. Image labeling
- C. Clustering
- D. House price prediction

Answer: C

13.

Reinforcement learning is based on:

- A. Fixed labeled datasets
- B. Explicit rules
- C. Rewards from sequences of actions
- D. Feature extraction only

Answer: C

14.

In reinforcement learning, the agent's goal is to:

- A. Minimize error on labeled data
- B. Maximize cumulative reward over time
- C. Discover hidden structure
- D. Reduce dimensionality

Answer: B

Part 4: ML Tasks and Concepts

15.

In Tom Mitchell's definition, learning is defined by improvement in:

- A. Runtime
- B. Performance P on task T with experience E
- C. Memory usage
- D. Code complexity

Answer: B

16.

In text classification, the performance measure P is commonly:

- A. Training time
- B. Memory usage
- C. Percentage of correctly classified texts
- D. Model size

Answer: C

17.

The "target function" $f: X \rightarrow Y$ represents:

- A. The learned hypothesis
- B. The true unknown mapping in reality
- C. The training dataset
- D. The optimization algorithm

Answer: B

18.

The hypothesis $g: X \rightarrow Y$ represents:

- A. The ground truth
- B. The validation set
- C. The model chosen by the learning algorithm
- D. The loss function

Answer: C

Part 5: Building an ML System

19.

Which step comes first in building a machine learning system?

- A. Modeling
- B. Validation
- C. Data collection
- D. Feature extraction

Answer: C

20.

Feature representation refers to:

- A. Choosing labels
- B. Encoding raw data into numerical features
- C. Running classifiers
- D. Testing performance

Answer: B

21.

The purpose of model estimation (training) is to:

- A. Collect more data
- B. Choose hyperparameters
- C. Find parameters that best fit the training data
- D. Evaluate performance

Answer: C

22.

Validation is mainly used to:

- A. Train the final model
- B. Tune hyperparameters and compare models
- C. Increase dataset size
- D. Replace testing

Answer: B



Part 6: Image Classification & Features

23.

A training set is best described as:

- A. Data never seen by the model
- B. Data used to evaluate final performance
- C. Data used to learn model parameters
- D. Data used only for visualization

Answer: C

24.

A good feature representation should be:

- A. Highly correlated
- B. Sensitive to noise
- C. Robust and discriminative
- D. Dependent on scale

Answer: C

25.

Feature independence is important because:

- A. It increases redundancy
- B. Correlated features add unnecessary information
- C. It reduces accuracy
- D. It simplifies labeling

Answer: B



Part 7: Classifiers & KNN

26.

A classifier is a function that:

- A. Compresses data
- B. Maps features to labels
- C. Generates features
- D. Collects data

Answer: B

27.

Which of the following is NOT a model class?

- A. Decision Trees
- B. Support Vector Machines
- C. Compiler
- D. Neural Networks

Answer: C

28.

The Nearest Neighbor classifier works by:

- A. Learning explicit parameters
- B. Finding the closest training example
- C. Optimizing a loss function
- D. Reducing dimensionality

Answer: B

29.

In K-Nearest Neighbor (KNN), increasing K generally makes the model:

- A. More sensitive to noise
- B. Less smooth
- C. More stable but potentially biased
- D. Unsupervised

Answer: C

Part 8: Hyperparameters & Validation

30.

Why should hyperparameters NOT be tuned using the test set?

- A. It increases training time
- B. The test set must represent unseen data for final evaluation
- C. Hyperparameters cannot be changed
- D. Test data is unlabeled

Answer: B